CHECKPOINT: MANAGING OPEN SSL

The aim of this checkpoint is to learn how to create and manage *open SSL certificates.*

In the process of learning and given some certain instructions, I will outline the processes below.

Launch Kali Linux- open terminal - check the open SSL version

create directory to save keys : mkdir keys (creating a file named keys)

generate asymmetric encryption keys: openssl genrsa -out corp.gomycode.com.key 2048

run ls to verify the creation of the file

show the certification : cat corp.gomycode.com.key

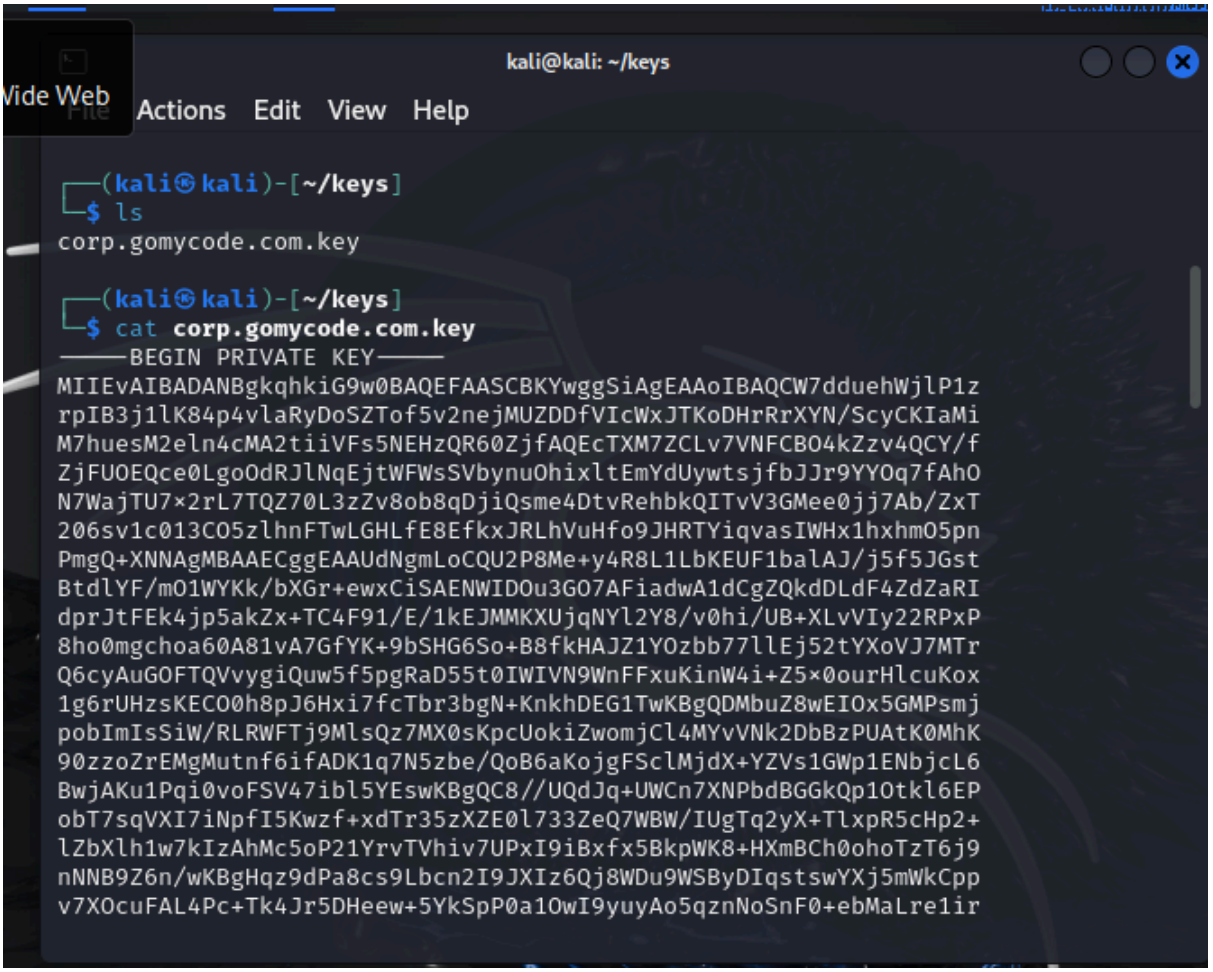
extract public key : openssl rsa -in corp.gomycode.com.key -pubout -out corp.gomycode.com_public.key

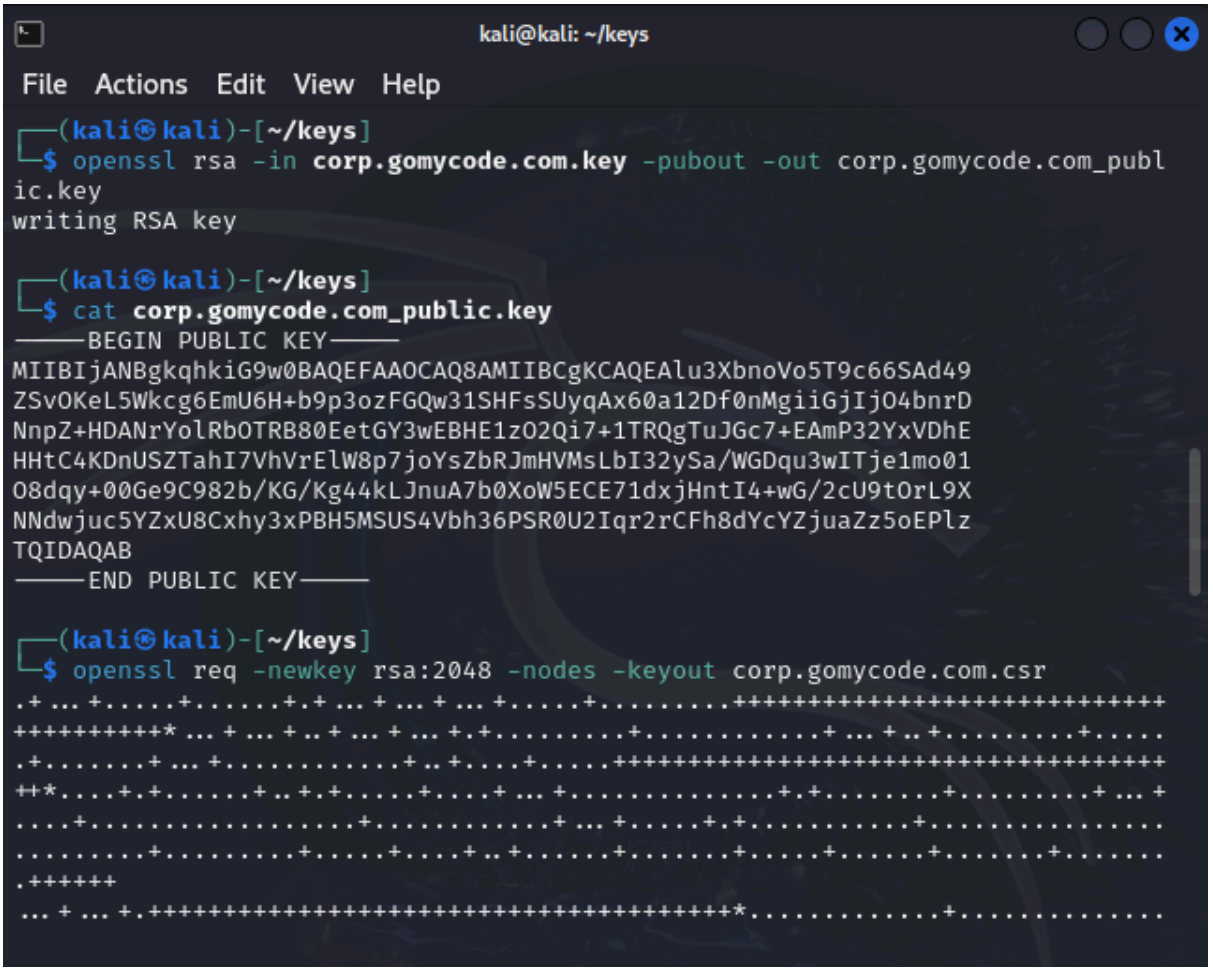
Generate a certificate signing request. Type the following command, then press ENTER

After then run openssl req -new -key corp.gomycode.com.key -out corp.gomycode.com.csr

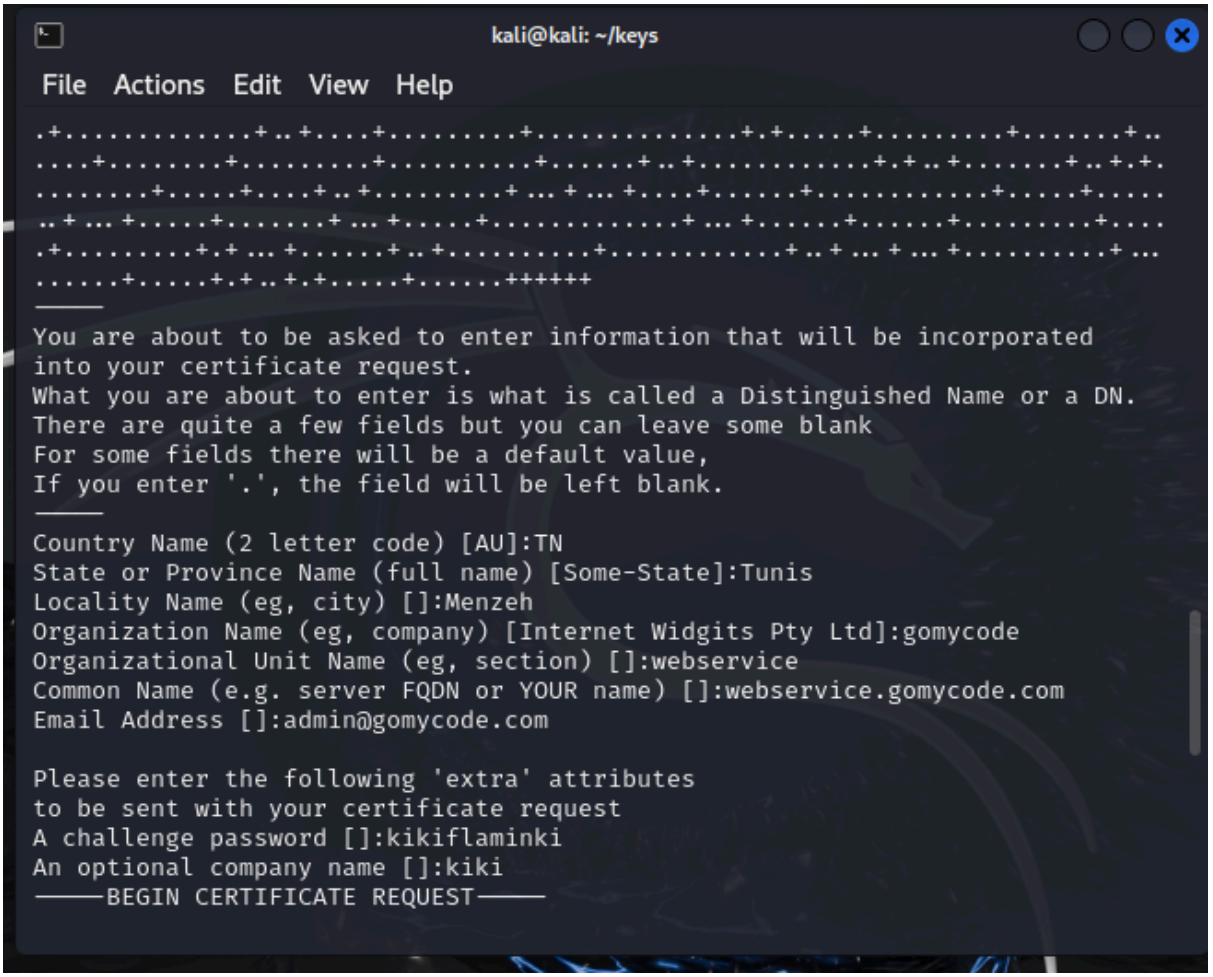
SEE IMAGE BELOW




kali@kali: ~/keys
Actions Edit View Help
(kali㉿kali)-[~/keys]
$ ls
corp.gomycode.com.key
(kali㉿kali)-[~/keys]
$ cat corp.gomycode.com.key
-----BEGIN PRIVATE KEY-----
MIIEvAIBADANBgkqhkiG9w0BAQEFAASCBKYwggSiAgEAAoIBAQCW7dduehWjlP1z
rpIB3j1lK84p4vlaRyDoSZTof5v2nejMUZDDfVIcWxJTKoDHrRrXYN/ScyCKIaMi
M7huesM2eln4cMA2tiiVFs5NEHzQR60ZjfAQEcTXM7ZCLv7VNFCBO4kZzv4QCY/f
ZjFUOEQce0LgoOdRJlNqEjtWFWsSVbynuOhixltEmYdUywtsjfbJJr9YYOq7fAhO
N7WajTU7×2rL7TQZ70L3zZv8ob8qDjiQsme4DtvRehbkQITvV3GMee0jj7Ab/ZxT
206sv1c013CO5zlhnFTwLGHLfE8EfkxJRLhVuHfo9JHRTYiqvasIWHx1hxhmO5pn
PmgQ+XNNAgMBAAECggEAAUdNgmLoCQU2P8Me+y4R8L1LbKEUF1balAJ/j5f5JGst
BtdlYF/mO1WYKk/bXGr+ewxCiSAENWIDOu3GO7AFiadwA1dCgZQkdDLdF4ZdZaRI
dprJtFEk4jp5akZx+TC4F91/E/1kEJMMKXUjqNYl2Y8/v0hi/UB+XLvVIy22RPxP
8ho0mgchoa60A81vA7GfYK+9bSHG6So+B8fkHAJZ1YOzbb77llEj52tYXoVJ7MTr
Q6cyAuGOFTQVvygiQuw5f5pgRaD55t0IWIVN9WnFFxuKinW4i+Z5×0ourHlcuKox
1g6rUHzsKECO0h8pJ6Hxi7fcTbr3bgN+KnkhDEG1TwKBgQDMbuZ8wEIOx5GMPsmj
pobImIsSiW/RLRWFTj9MlsQz7MX0sKpcUokiZwomjCl4MYvVNk2DbBzPUAtK0MhK
90zzoZrEMgMutnf6ifADK1q7N5zbe/QoB6aKojgFSclMjdX+YZVs1GWp1ENbjcL6
BwjAKu1Pqi0voFSV47ibl5YEswKBgQC8//UQdJq+UWCn7XNPbdBGGkQp1Otkl6EP
obT7sqVXI7iNpfI5Kwzf+xdTr35zXZE0l733ZeQ7WBW/IUgTq2yX+TlxpR5cHp2+
lZbXlh1w7kIzAhMc5oP21YrvTVhiv7UPxI9iBxfx5BkpWK8+HXmBCh0ohoTzT6j9
nNNB9Z6n/wKBgHqz9dPa8cs9Lbcn2I9JXIz6Qj8WDu9WSByDIqstswYXj5mWkCpp
v7XOcuFAL4Pc+Tk4Jr5DHeew+5YkSpP0a1OwI9yuyAo5qznNoSnF0+ebMaLre1ir



After running the command it generated a certificate signing request in which I filled out a form. **SEE IMAGE BELOW**



Next is:

verify the certificate request : openssl req -text -in corp.gomycode.com.csr -noout -verify

Ran the code to generate a sel-signed certificate

openssl req -newkey rsa:2048 -nodes -keyout corp.gomycode.com.key -x509 -days 365 -out corp.gomycode.com.crt

openssl pkcs12 -export -name "corp.gomycode.com" -out corp.gomycode.com.pfx -inkey corp.gomycode.com.key -in corp.gomycode.com.crt

See image below:

```
┌──(kali㉿kali)-[~/keys]
└─$ openssl req -newkey rsa:2048 -nodes -keyout corp.gomycode.com.key -x509 -
days 365 -out corp.gomycode.com.crt

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You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
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Country Name (2 letter code) [AU]:kiki
String too long, must be at most 2 bytes long
Country Name (2 letter code) [AU]:Nigeria
String too long, must be at most 2 bytes long
Country Name (2 letter code) [AU]:NG
State or Province Name (full name) [Some-State]:lagos
Locality Name (eg, city) []:lagos
Organization Name (eg, company) [Internet Widgits Pty Ltd]:kiki flaminki
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:Blessing
Email Address []:akidavies1@gmail.com
```

```
┌──(kali㉿kali)-[~/keys]
└─$ openssl pkcs12 -export -name "corp.gomycode.com" -out corp.gomycode.com.p
fx -inkey corp.gomycode.com.key -in corp.gomycode.com.crt
Enter Export Password:
Verifying - Enter Export Password:

┌──(kali㉿kali)-[~/keys]
└─$
```

THANK YOU.